# Adilkhan Salkimbayev

Applied Mathematics & Scientific Machine Learning Researcher
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## EDUCATION

**Kazakh-British Technical University (KBTU)**

B.Sc. in Automation and Control | GPA: 3.58 | Aug 2023 – July 2027 (Expected)

- **Relevant Coursework:** Linear Algebra for Engineers, Differential Equations, Calculus of Complex Variables
- **Military Department:** Sergeant of the Reserve (Completed June 2025).

## PUBLICATIONS AND PREPRINTS

- **A. Salkimbayev.** "Certifiable Approximation of Model Predictive Control Laws: A Shape-Constrained Polynomial Read-Out." *Under Review, Journal of Process Control.*
- **A. Salkimbayev, K.S. Haider.** "Robust Adaptive Kolmogorov-Arnold Neural Control." *Under Review, International Journal of Adaptive Control and Signal Processing.*

## RESEARCH EXPERIENCE

**Independent Undergraduate Researcher | Kazakh-British Technical University**

*Functional-Analytic Control Architecture & PDEs* (Current)

- Developing a novel functional-analytic control architecture drawing upon operator theory and methodologies from Brezis, Evans and Stein.
- Translating control frameworks into rigorous mathematical formulations applicable to the study of partial differential equations and fluid dynamics.

*Adaptive Control & Nonlinear Dynamical Systems* (Jan 2026 – Jul 2026)

- Developed a learning-based adaptive control architecture combining Recursive Least Squares with symbolic Kolmogorov-Arnold Networks (KANs).
- Derived rigorous Lyapunov stability guarantees (ISS / UUB) over the physically realizable operating envelope, with a certificate that requires no persistent excitation.
- Evaluated the algorithm on a quadruple-tank benchmark via a 50-seed Monte-Carlo campaign in Python, demonstrating a branch-free per-step runtime bounded a priori and a 7% aggregate tracking improvement over LTV-MPC.

*Constant-Time Neural Control for Embedded Systems* (Nov 2025 – Apr 2026)

- Designed a symbolic KAN architecture as a computationally constrained alternative to Linear Time-Varying Model Predictive Control (MPC) for non-minimum phase systems.
- Executed Hardware-in-the-Loop (HIL) methodology on an STM32H7 microcontroller, achieving a runtime reduction of up to 5 orders of magnitude (OOM).

## ADVANCED INDEPENDENT STUDY

Self-directed graduate sequence in analysis and PDE, directed toward the rigorous study of fluid dynamics:

- **Completed:** *Introduction to Real Analysis* (William F. Trench); *Functional Analysis, Sobolev Spaces and Partial Differential Equations* (Haim Brezis), Chs. 1–9 — including semigroup theory (Hille–Yosida) and the variational formulation of elliptic boundary-value problems; *Partial Differential Equations* (Lawrence C. Evans), Chs. 5–6 — Sobolev spaces and second-order elliptic theory.

- **In progress:** *Real Analysis: Measure Theory, Integration, and Hilbert Spaces* (Elias M. Stein & Rami Shakarchi), Chs. 1–3.
- **Planned (Fall 2026):** *Measure Theory and Fine Properties of Functions* (Lawrence C. Evans & Ronald F. Gariepy); *Singular Integrals and Differentiability Properties of Functions* (Elias M. Stein).

## TECHNICAL & COMPUTATIONAL SKILLS

**Mathematics & Control Systems:**
Operator theory, partial differential equations, semigroup theory, Lyapunov stability analysis, adaptive control, nonlinear control systems

**Scientific Machine Learning:**
Kolmogorov-Arnold Networks, system identification, embedded SciML.

**Programming & Hardware:** Python (NumPy, SciPy, matplotlib, pandas, PyTorch), C/C++, STM32 CubeIDE, LaTeX (VSCode).